Head & Neck Osteology: The 4 Normas & Skull Introduction (Set 1)

1. Which suture forms the junction between the two parietal bones along the midline of the skull vault in Norma Verticalis?
a. Sagittal suture
b. Coronal suture
c. Lambdoid suture
d. Metopic suture
e. Squamous suture
2. The coronal suture represents the structural line of articulation between which of the following bones?
a. The two parietal bones
b. The frontal bone and the two parietal bones
c. The parietal bone and occipital bone
d. The frontal bone and zygomatic bone
e. The temporal bone and parietal bone
3. The point of intersection between the sagittal suture and the coronal suture is an essential anthropometric landmark known as the:
a. Lambda
b. Bregma
c. Pterion
d. Asterion
e. Inion
4. In a newborn skull, the bregma is represented by a membrane-covered gap that allows for brain expansion. This structure is the:
a. Posterior fontanelle
b. Sphenoidal fontanelle
c. Mastoid fontanelle
d. Anterior fontanelle
e. Metopic fontanelle
5. Clinical box details note that the anterior fontanelle normally undergoes complete bony closure by which postnatal age?
a. 2 to 3 months
b. 6 months
c. 12 months
d. 18 to 24 months
e. 36 months
6. The posterior fontanelle sits at the intersection of the sagittal and lambdoid sutures (Lambda) and normally closes by which age?
a. 12 months
b. 2 to 3 months
c. 18 months
d. 9 months
e. 24 months

7. A pair of small openings located on either side of the sagittal suture near the back of Norma Verticalis transmits an emissary vein to the superior sagittal sinus. These are the:
a. Parietal foramina
b. Mastoid foramina
c. Foramina lacerum
d. Foramina spinosum
e. Jugular foramina
8. The precise unossified segment of the sagittal suture situated between the two parietal foramina is named the:
a. Bregma
b. Obelion
c. Lambda
d. Pterion
e. Asterion
9. Looking at Norma Frontalis, which prominent bilateral opening in the frontal bone transmits the supraorbital nerve and vessels to the forehead?
a. Supraorbital notch or foramen
b. Infraorbital foramen
c. Mental foramen
d. Zygomaticofacial foramen
e. Optic canal
10. The prominent midline landmark at the root of the nose, where the internasal suture meets the frontonasal suture, is the:
a. Glabella
b. Inion
c. Nasion
d. Bregma
e. Pterion
11. The smooth, slightly elevated area on the frontal bone situated immediately superior to the nasion and between the two superciliary arches is the:
a. Inion
b. Glabella
c. Lambda
d. Obelion
e. Asterion
12. The infraorbital foramen is an important local anesthesia landmark located on the anterior surface of the maxilla. It transmits the infraorbital nerve, which is a direct branch of the:
a. Ophthalmic nerve (V1)
b. Mandibular nerve (V3)
c. Facial nerve (CN VII)
d. Maxillary nerve (V2)
e. Oculomotor nerve (CN III)
13. The bony opening on the anterior surface of the maxilla that leads directly into the nasal cavity is called the:
a. Piriform aperture
b. Choana

c. Superior orbital fissure
d. Pterygoid canal
e. Foramen rotundum

14. Norma Occipitalis features a prominent midline projection on the external surface of the occipital bone. This structural landmark is the:

a. Internal occipital protuberance
b. External occipital protuberance
c. Superior nuchal line
d. Jugular tubercle
e. Pharyngeal tubercle

15. The most prominent, central point located on the external occipital protuberance is designated as the:

a. Nasion
b. Bregma
c. Inion
d. Pterion
e. Asterion

16. Which H-shaped sutural junction on the aspect of Norma Lateralis is formed by the articulation of the frontal, parietal, temporal, and sphenoid bones?

a. Asterion
b. Lambda
c. Bregma
d. Obelion
e. Pterion

17. The pterion is a high-yield clinical region because it overlies the anterior division of which major blood vessel, whose rupture causes an epidural hematoma?

a. Internal carotid artery
b. Middle cerebral artery
c. Middle meningeal artery
d. Superficial temporal artery
e. Maxillary artery

18. The horizontal bony bar on Norma Lateralis that is formed by the temporal process of the zygomatic bone and the zygomatic process of the temporal bone is the:

a. Temporal line
b. Zygomatic arch
c. Styloid process
d. Mastoid process
e. Pterygoid plate

19. The temporal lines sweep across the lateral aspect of the skull skull. The area situated inferior to these lines, bounded laterally by the zygomatic arch, forms the:

a. Infratemporal fossa
b. Pterygopalatine fossa
c. Mandibular fossa
d. Temporal fossa
e. Cranial vault

20. The junction of the lambdoid, parietomastoid, and occipitomastoid sutures on the posterolateral aspect of the skull is the:

a. Pterion
b. Bregma
c. Nasion
d. Asterion
e. Inion

21. According to the B.D. Chaurasia clinical summary points, the asterion overlies the location of which intracranial venous channel?

a. Cavernous sinus
b. Sigmoid sinus
c. Superior sagittal sinus
d. Straight sinus
e. Inferior petrosal sinus

22. Fractures of the skull can occur along various lines. A fracture where a fragment of bone is driven inward, directly compressing or tearing the underlying brain tissue, is a:

a. Linear fracture
b. Fissured fracture
c. Depressed fracture
d. Punctured fracture
e. Counter-coup fracture

23. When an impact to one side of the skull causes a fracture on the completely opposite side of the skull, the mechanical phenomenon is termed a:

a. Coup injury
b. Direct fracture
c. Depressed injury
d. Counter-coup (contrecoup) fracture
e. Diastatic fracture

24. Premature pathological closure of the cranial sutures in an infant leads to a deformed skull and increased intracranial pressure. This clinical condition is:

a. Hydrocephalus
b. Craniosynostosis (Craniostenosis)
c. Anencephaly
d. Microcephaly
e. Acromegaly

25. Premature closure of the sagittal suture leads to a skull that is elongated in the anteroposterior diameter and narrow transversely, a specific shape called:

a. Oxycephaly
b. Plagiocephaly
c. Brachycephaly
d. Trigonocephaly
e. Scaphocephaly (Dolichocephaly)

26. Premature closure of the coronal suture results in a short, broad skull configuration known clinically as:

a. Scaphocephaly
b. Brachycephaly
c. Oxycephaly
d. Plagiocephaly
e. Trigonocephaly

27. The zygomaticofacial foramen is located on the lateral surface of the zygomatic bone. It transmits a nerve that provides sensory supply to the skin of the cheek, which arises from:
a. Maxillary nerve (V2)
b. Ophthalmic nerve (V1)
c. Mandibular nerve (V3)
d. Facial nerve (CN VII)
e. Infraorbital nerve
28. The external acoustic meatus is located on Norma Lateralis. The sharp bony margin forming its anterior and inferior border is derived from which part of the temporal bone?
a. Squamous part
b. Petrous part
c. Mastoid part
d. Tympanic part
e. Styloid process
29. The mastoid process contains air cells that communicate with the middle ear cavity. Inflammation can track here from the middle ear, a condition called:
a. Otitis externa
b. Mastoiditis
c. Sinusitis
d. Labyrinthitis
e. Parotitis
30. The mastoid foramen is located near the posterior border of the mastoid part of the temporal bone. It transmits an emissary vein that connects the scalp veins directly to the:
a. Cavernous sinus
b. Superior sagittal sinus
c. Sigmoid sinus
d. Inferior sagittal sinus
e. Transverse sinus
31. The superciliary arches are prominent bony ridges located superior to the orbital margins in Norma Frontalis. They are typically much more developed and prominent in:
a. Adult females
b. Infants and newborns
c. Adult males
d. Elderly individuals only
e. Congenital hydrocephalus cases
32. The anterior nasal spine is a sharp, midline bony projection located at the inferior margin of the piriform aperture, formed by the union of the two:
a. Nasal bones
b. Palatine bones
c. Maxillae
d. Zygomatic bones
e. Conchae
33. The vertical suture that occasionally persists between the two halves of the frontal bone in a small percentage of adult skulls is the:

a. Sagittal suture
b. Coronal suture
c. Metopic (frontal) suture
d. Lambdoid suture
e. Squamous suture

34. The temporal fossa contains the temporalis muscle. The floor of this fossa is formed in part by the greater wing of which bone?

a. Occipital bone
b. Ethmoid bone
c. Sphenoid bone
d. Frontal bone
e. Maxilla

35. A fracture passing through the floor of the anterior cranial fossa can tear the overlying meninges and mucous membranes, causing cerebrospinal fluid to leak into the nose. This is:

a. CSF otorrhea
b. Epistaxis
c. Hematemesis
d. Hemoptysis
e. CSF rhinorrhea

36. Laceration of the middle meningeal artery at the pterion causes a collection of blood that displays which characteristic configuration on a head CT scan?

a. Crescent-shaped (concave)
b. Biconvex, lens-shaped (lentiform)
c. Diffuse layering along the sulci
d. Intraventricular flooding
e. Circular ring-enhancing mass

37. The highest nuchal lines, when present, arise from the external occipital protuberance and extend laterally superior to the:

a. Inferior nuchal lines
b. Superior nuchal lines
c. Temporal lines
d. Mastoid processes
e. Occipital condyles

38. The point where the frontonasal suture intersects with the internasal suture is named the:

a. Glabella
b. Nasion
c. Inion
d. Lambda
e. Bregma

39. The bony orbital margin is formed by multiple bones. The superior orbital margin is formed entirely by the:

a. Zygomatic bone
b. Maxilla
c. Frontal bone
d. Sphenoid bone
e. Ethmoid bone

40. The structural stability of the face depends on vertical and horizontal buttresses. The main vertical buttress that transmits forces from the maxillary teeth to the skull base is the:
a. Zygomatic arch
b. Pterygoid plate column
c. Nasomaxillary (nasofrontal) buttress
d. Styloid process
e. Occipital condyle axis

Answer Key Table (Set 1)

Q	Q	Q	Q	Q

1-8	9-16	17-24	25-32	33-40
a	a	c	e	c
b	c	d	b	c
b	b	d	b	e
d	a	b	e	c
d	d	e	a	b
b	e	d	d	c
a	e	a	c	c
a	e	c	c	c